

genetics



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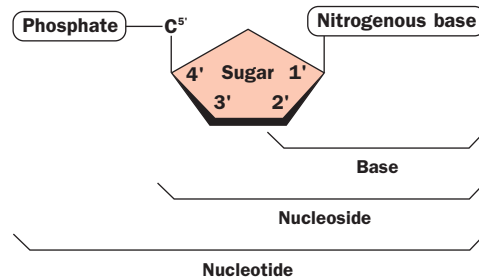
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For Your Reference

The following section provides a group of diagrams and illustrations applicable to many entries in this encyclopedia. The molecular structures of DNA and RNA are provided in detail in several different formats, to help the student understand the structures and visualize how these molecules combine and interact. The full set of human chromosomes are presented diagrammatically, each of which is shown with a representative few of the hundreds or thousands of genes it carries.

NUCLEOTIDE STRUCTURE



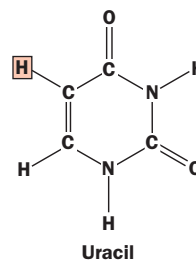
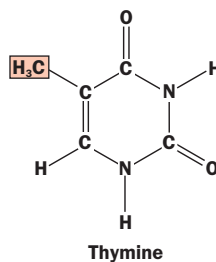
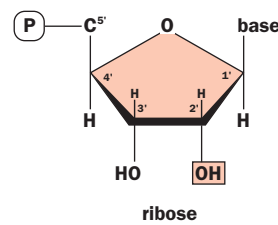
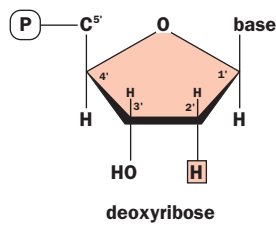
Sample naming conventions for each structure:

Adenine

Adenosine

Adenosine monophosphate

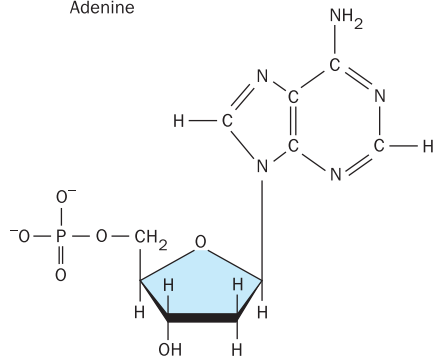
DNA VS. RNA



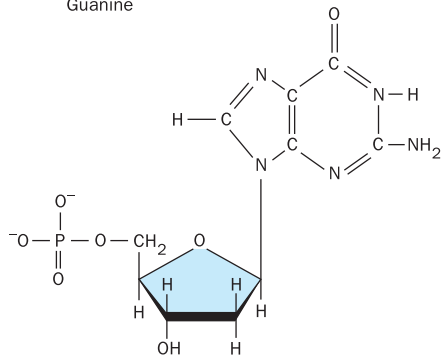
NUCLEOTIDE STRUCTURES

Purine-containing DNA nucleotides

Adenine

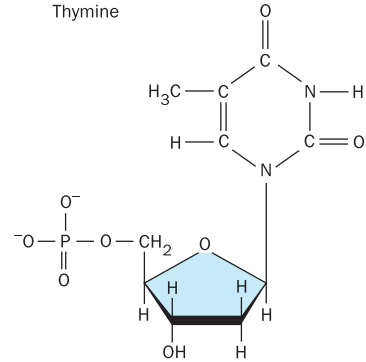


Guanine

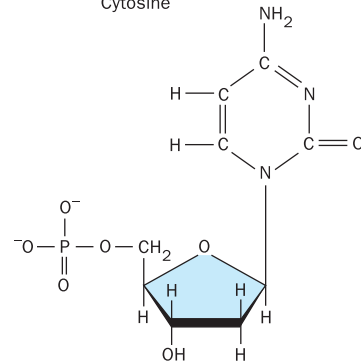


Pyrimidine-containing DNA nucleotides

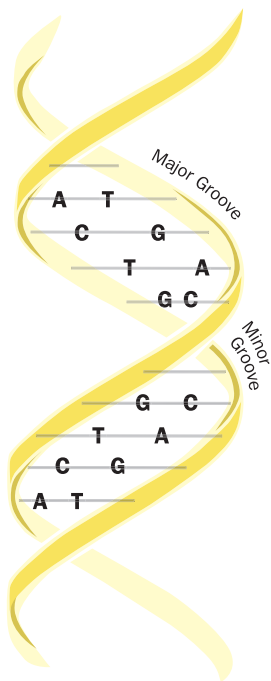
Thymine



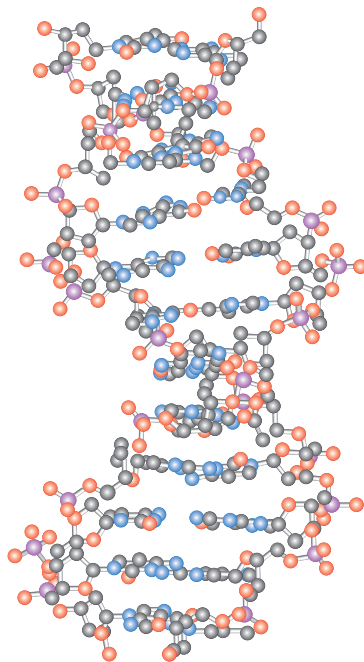
Cytosine



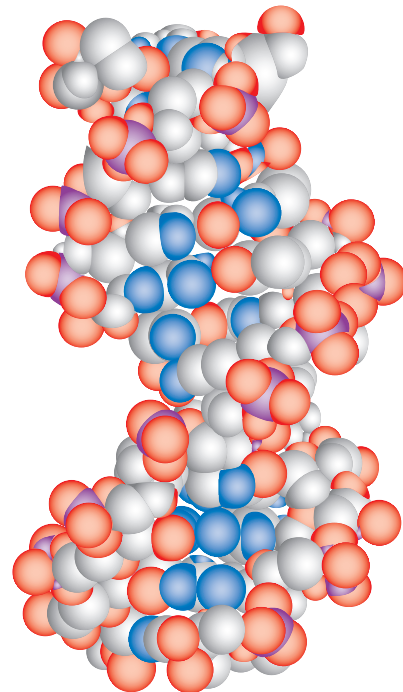
CANONICAL B-DNA DOUBLE HELIX



Ribbon model

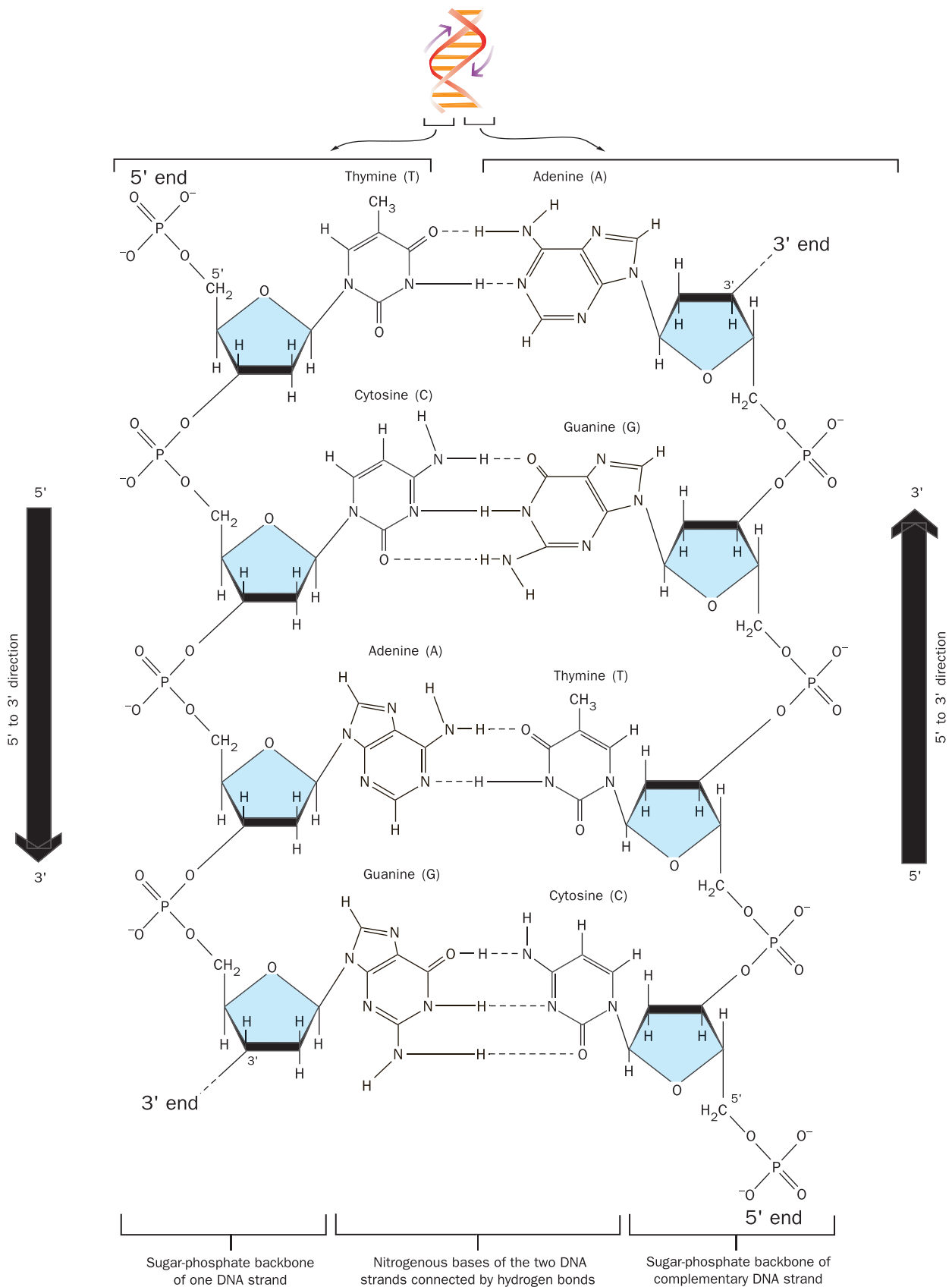


Ball-and-stick model

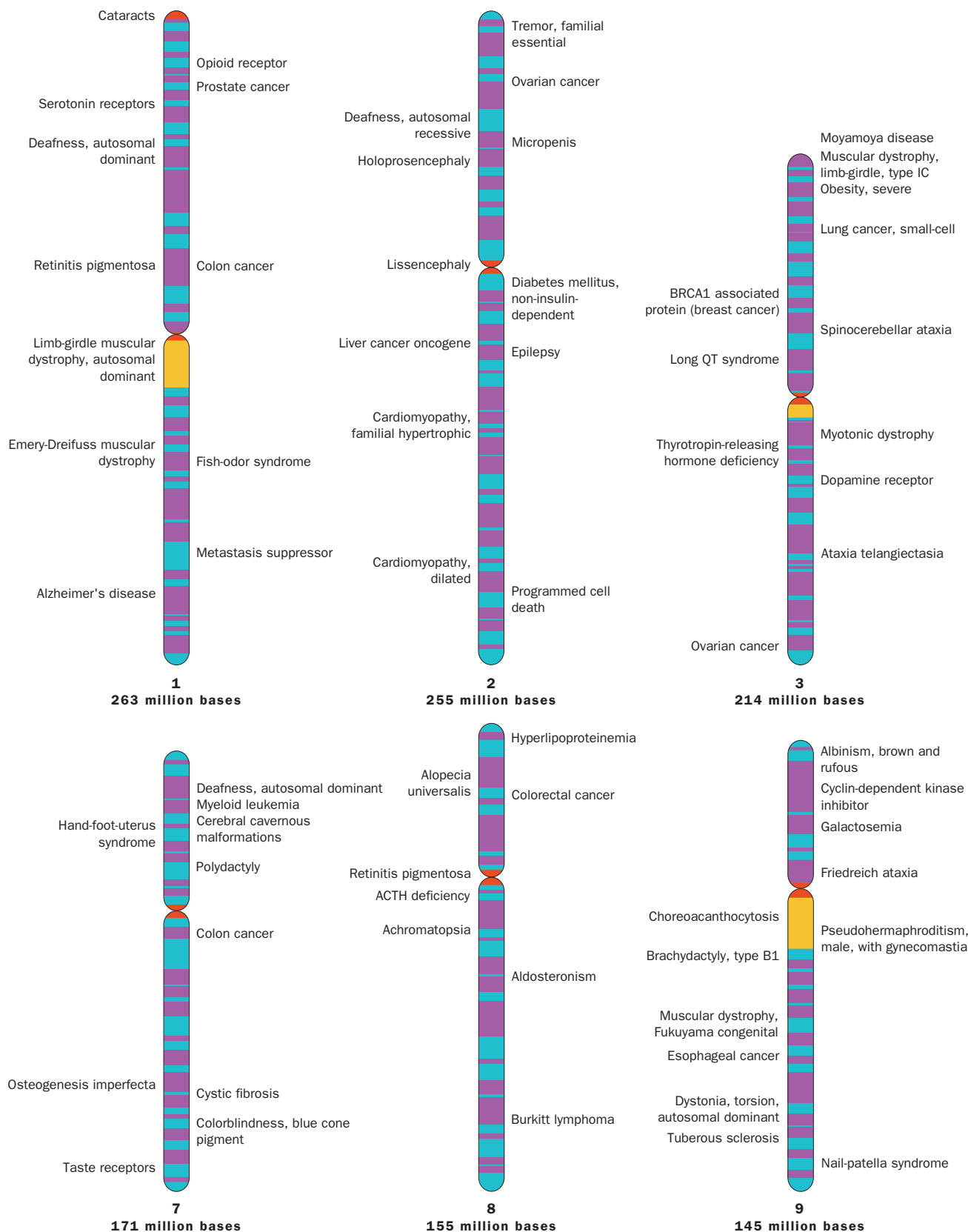


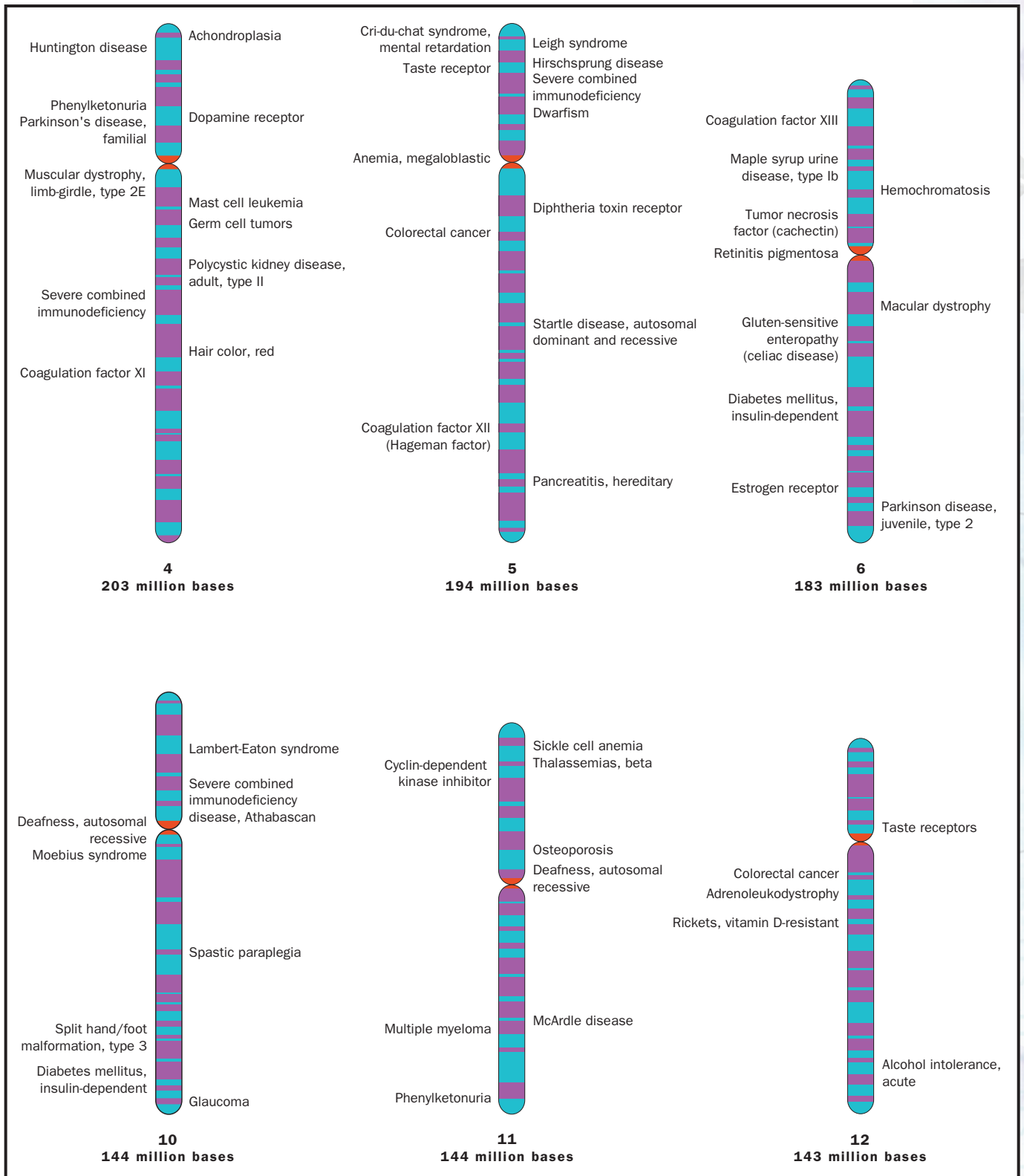
Space-filling model

DNA NUCLEOTIDES PAIR UP ACROSS THE DOUBLE HELIX; THE TWO STRANDS RUN ANTI-PARALLEL

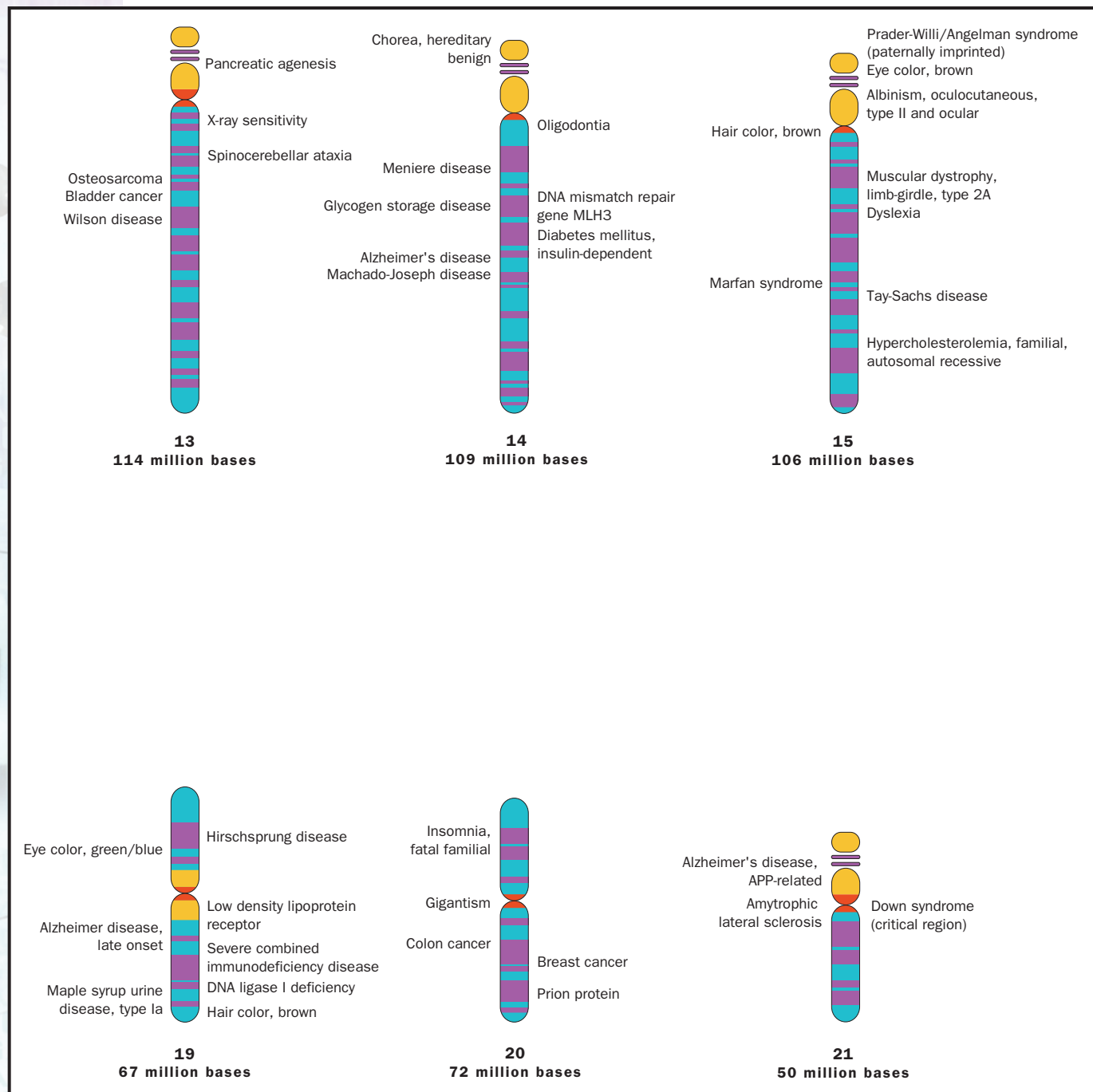


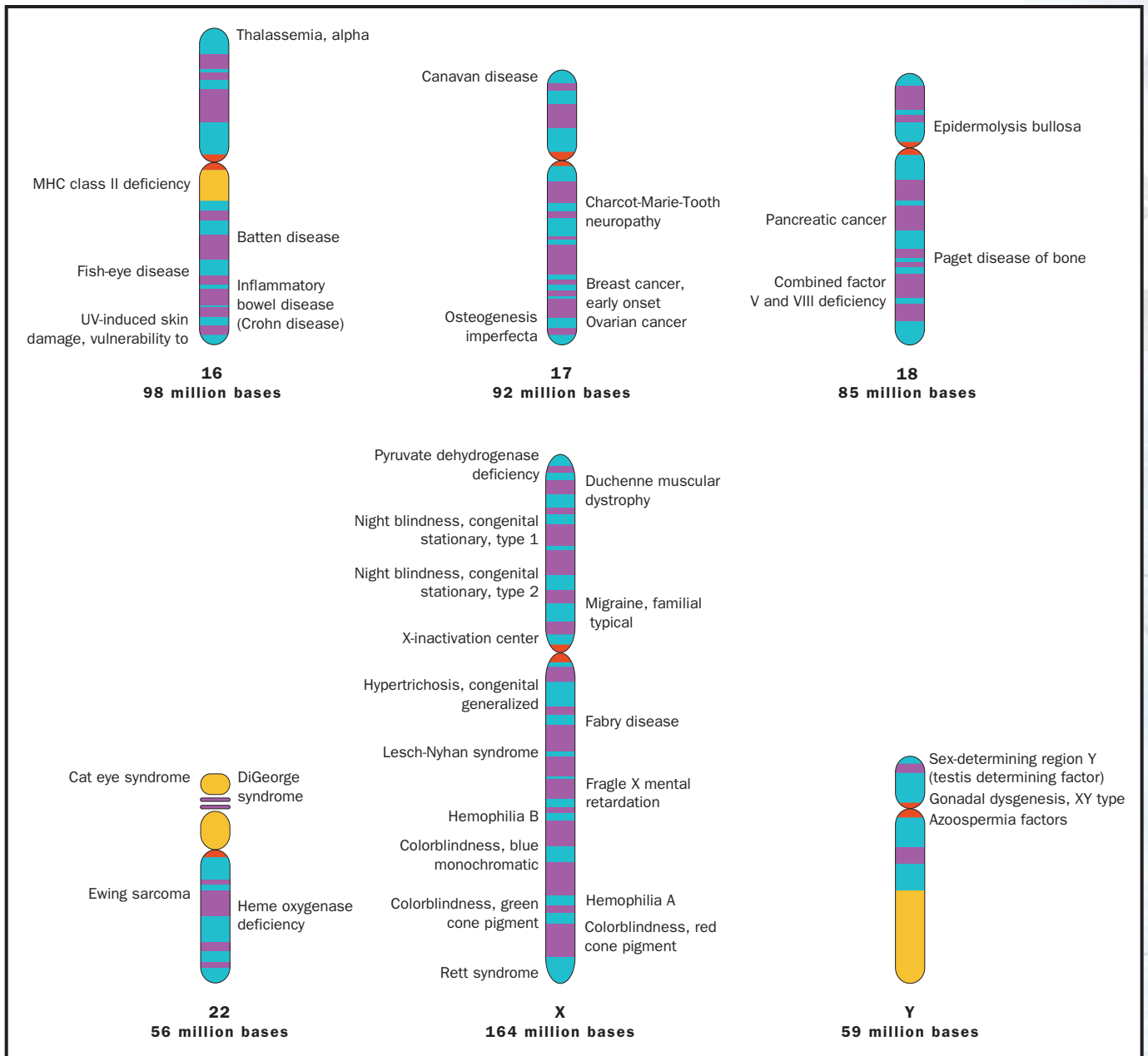
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Karyotype *See Chromosomal Banding; Nomenclature*

Knockout *See Gene Targeting*

Laboratory Technician

The technician in a molecular biology laboratory is a resourceful scientist who specializes in the various experimental techniques critical to the mission of the laboratory. The work offers many rewards beyond the financial ones. For example, one of the rewards of working as a laboratory technician lies in being part of a team dedicated to scientific discovery. Another rewarding aspect of the laboratory technician position is the need for continual learning, as new scientific techniques replace older ones.

Skills of the Laboratory Technician

Technicians must possess a variety of skills, depending on the work being done in the laboratory in which they work. For example, a technician in a laboratory that studies human genetic **polymorphisms** will be skilled in the techniques of DNA isolation and DNA sequencing. DNA may be isolated from cultured human cells, which the technician would grow, or from tissue biopsies or blood samples. The technician must exercise great care to prevent accidental contamination of the samples.

Following (chemical) extraction of DNA from the material, the technician will **amplify** the region of the gene under investigation in an enzyme-catalyzed DNA sequencing reaction. The reaction products, which are pieces of DNA of various lengths, are loaded by the technician onto a thin gel and separated from one another according to size by applying an electric current to the gel. A computer-controlled laser excites fluorescent dye molecules that the technician has chemically attached to the DNA, and a photodetector records the color. The laboratory technician will operate the DNA sequencing machine, supervise the electronic data collection, and, in general, assure the accuracy of the DNA sequence obtained. Good communication skills are an important part of a technician's qualifications, as is the ability to follow (and give) instructions correctly.

polymorphisms DNA sequence variants

amplify produce many copies of

K

L

In 2000, at Genetic ID, the nation's largest genetic testing lab in Fairfield, Iowa, a technician uses the triple check method to narrow results.



In a protein structure laboratory, on the other hand, the technician must have expertise in protein purification. Since relatively large amounts of protein may be required for structure determination, the gene encoding the protein may be placed into bacteria using recombinant DNA techniques. The bacteria can then be induced to produce (express) significant quantities of the protein. The technician is in charge of growing the bacteria and seeing whether the protein is properly expressed. Next, the technician breaks open the bacterial cells by mechanical means, and purifies (separates) the protein of interest away from contaminating proteins and nucleic acids using a series of chromatographic techniques. The technician must know how to detect and quantify proteins and **enzymes**.

enzymes proteins that control a reaction in a cell

Qualifications and Compensation

Often, a laboratory technician can demonstrate expertise in a wide variety of experimental techniques. This type of employee is highly sought after by the pharmaceutical and biotechnology industries. It is generally expected that a technician in a molecular biology work setting will have a bachelor's or master's degree in biology, biochemistry, or chemistry.

While many employers are willing to provide on-the-job training for laboratory technicians, a well-qualified technician will have at least some experience with techniques commonly used in biochemistry, genetics, and cell biology.

A laboratory technician who has just received a bachelor's or master's degree may receive an annual salary of \$18,000 to \$36,000 or more, depending on the employer (industry, academia, or government), the geographical location, and the supply of and demand for qualified technicians. Salaries increase with experience, technical expertise, and responsibility, particularly in industry, where many opportunities exist for the laboratory technician to climb the career ladder.

Samuel E. Bennett and Dale Mosbaugh

Legal Issues

The question of who owns tissues, DNA, and other biological materials raises numerous legal questions. One concern is that genetic information derived from someone's DNA sequences could be used to deny insurance coverage to people whose genes indicate that they have a disease or that they are at risk of contracting one.

Another concern is that the profits made by hospitals and transplant centers for transplantation procedures are unfair, as tissue donors and their families are typically not compensated, despite the fact that these donors often pay for the operations that provide the materials. There is a question of who should profit from such materials: those from whom the materials were originally derived or those who use the materials to treat other patients or conduct research.

In the criminal setting, genetic testing provides the opportunity to identify criminals. Through storage of **DNA** and DNA analysis data, old, unsolved cases can sometimes be resolved. DNA analysis is also useful for exonerating wrongly accused individuals, including those who have served significant jail time for crimes they did not commit. However, there is concern regarding the potential abuses of DNA data stored by law enforcement agencies. There is also concern that stored genetic material will be used to clone people. Additional concerns center on the safety and risks of genetically modified foods.

DNA deoxyribonucleic acid

Ownership of Tissues

The issue of ownership of tissues was addressed in California in the case *Moore v. Regents of the University of California*. Moore underwent treatment for leukemia at the University of California at Los Angeles Medical Center. His spleen was removed, and his cells were cultured without his consent. Eventually, he sued the medical center over the ownership of the cell line that was developed from his spleen cells.

The California Supreme Court refused to recognize Moore's ownership of the cell line, pointing to the investment the medical center made to develop it. The court did not place much weight on financial or other contributions Moore made to the development of the cell line.

The court indicated that recognizing a patient's right to own such cells would chill medical research, as scientists would be required to determine the originators of each cell culture they use. Because of the large number of cell cultures used, such a requirement would be burdensome and expensive, and it would potentially halt important research, the court said. The court also noted that researchers establishing cell lines are increasingly using contracts to clarify patent and ownership rights, though in Moore's case none was signed.

Criminal Law

DNA testing has proven to be a very valuable tool for convicting criminals, as well as for exonerating falsely accused individuals. The methods used to analyze DNA, as well as the implications of the results of such analyses, are still being standardized and are almost always questioned in court by at least one party, but they are becoming increasingly accepted and refined for use in criminal law.

Despite the usefulness of genetic testing, there are various concerns about privacy and the potential for discrimination. There are also some concerns about the consequences if insurance companies, employers, or other entities have access to such personal data.

Patenting Issues

Genetic material obtained from individuals is often used in developing patentable inventions. These patents are filed by the scientists who develop the materials and methods that utilize the genetic information. Cells obtained from a person with a rare disease, for example, might be used to develop tests to detect the disease as well as methods and materials for treatment. The patent rights are granted to the scientists who develop the tests, methods, and materials, rather than to the patient who was the source of the cells.

A patient might consider negotiating for an ownership interest in the cells. But very few patients are in a position to do so. They often are afraid that such negotiations would result in denial of treatment. Also, since it is illegal in the United States to sell organs and tissues, agreements involving ownership in such cases could be seen as falling afoul of the law. **SEE ALSO** DNA PROFILING; GENETIC COUNSELING; GENETIC DISCRIMINATION; GENETIC TESTING; GENETIC TESTING: ETHICAL ISSUES; PATENTING GENES; PRIVACY.

Kamrin T. MacKnight

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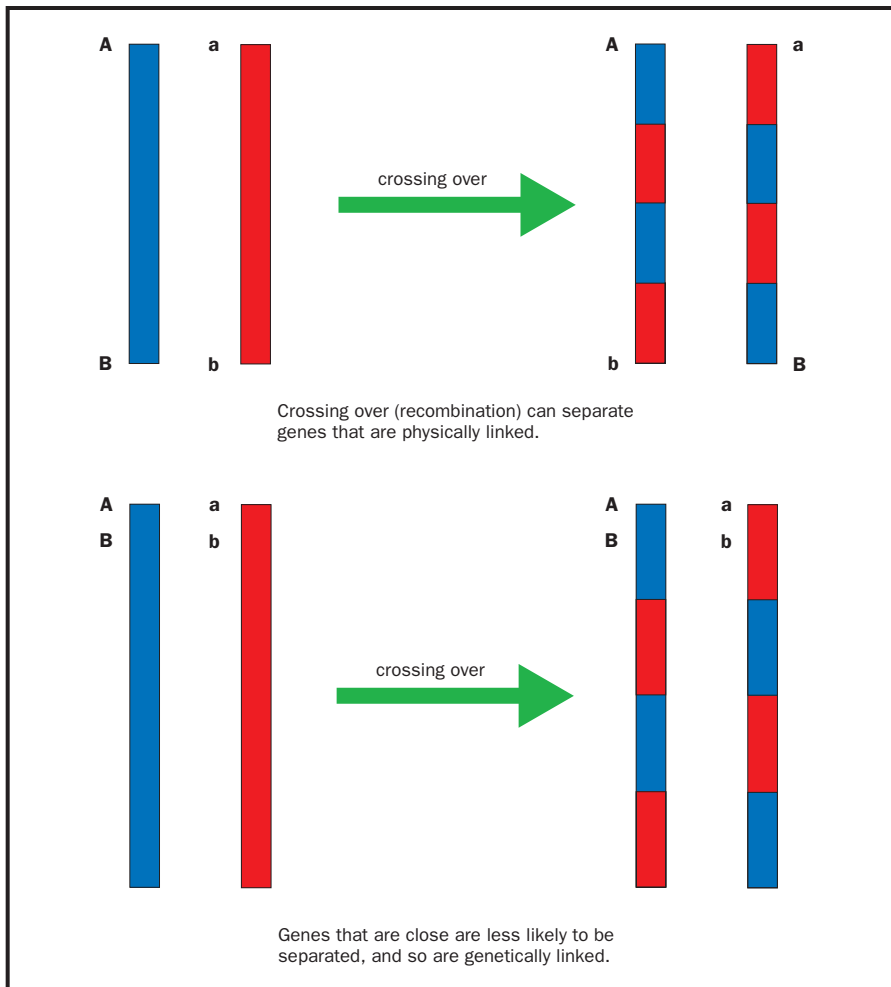
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Linkage and Recombination

Linkage refers to the association and co-inheritance of two DNA segments because they reside close together on the same chromosome. Recombination is the process by which they become separated during crossing over,



Physical linkage of genes simply means they are on the same chromosome. To be genetically linked, a pair of genes must be close enough that they are unlikely to be separated by crossing over.

which occurs during **meiosis**. The existence of linkage and the frequency of recombination allow chromosomes to be mapped to determine the relative positions and distances of the genes and other DNA sequences on them. Linkage analysis is also a key tool for discovering the location and ultimate identity of genes for inherited diseases.

Basic Concepts

Each individual inherits a complete set of twenty-three chromosomes from each parent, and chromosomes are therefore present in **homologous** pairs. The members of a pair carry the same set of genes at the same positions, or **loci**. The two genes at a particular locus may be identical, or slightly different. The different forms of a gene are called alleles.

Genes or loci can be linked either physically or genetically. Genes that are physically linked are on the same chromosome and are thus syntenic. Only syntenic genes can be genetically linked. Genes that are linked genetically are physically close enough to one another that they do not segregate independently during meiosis.

Understanding independent segregation is crucial to understanding linkage. Independent segregation was first discovered by Gregor Mendel, who found that, in pea plants, the different forms of two traits found in the

meiosis cell division that forms eggs or sperm

homologous possessing the same set of genes

loci sites on a chromosome (singular, locus)